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SAP Master Data Integration

Pieter Janssens, SAP Inside Track Belgium, 23/09/2023

MAKING AN IMPACT THAT MATTERS since 1845

Me in a nutshell

Pieter Janssens

Deloitte

Working in HR tech since 2012

Focus on SAP BTP, CF, integrations, extensions, UI5, CAP

@piejanssens

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SAP Master Data Integration

Introduction

SAP Master Data Integration

A Master Data Management solution running on SAP BT

- 1 central integration point
- 1 source of truth
- 1 lingo: SAP One Domain Model
- Out-of-the box integration scenario's to SAP solutions

- In addition
  - Available on trial accounts!
  - Pricing:
    - SAP cloud applications: 0 EUR
    - Connection with S/4HANA on-prem: ~50 EUR/month/GB storage
  - Integration:
    - SAP Cloud ALM for monitoring
    - SAP Master Data Governance for DQ
    - SAP CPI through a standard adapter
  - API's available
  - Extensible
  - Protection from dirty writes
  - Distribution model to manage filtering of data objects and fields
  - Data protection and privacy (e.g. read access logging)

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SAP Master Data Integration

1 Central integration point

2 SAP One Domain Model as lingua franca for exchanging data

3 Out of the box integrations provided to cloud solutions from SAP

Point-to-point integration → Central integration point

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SAP Master Data Integration

1

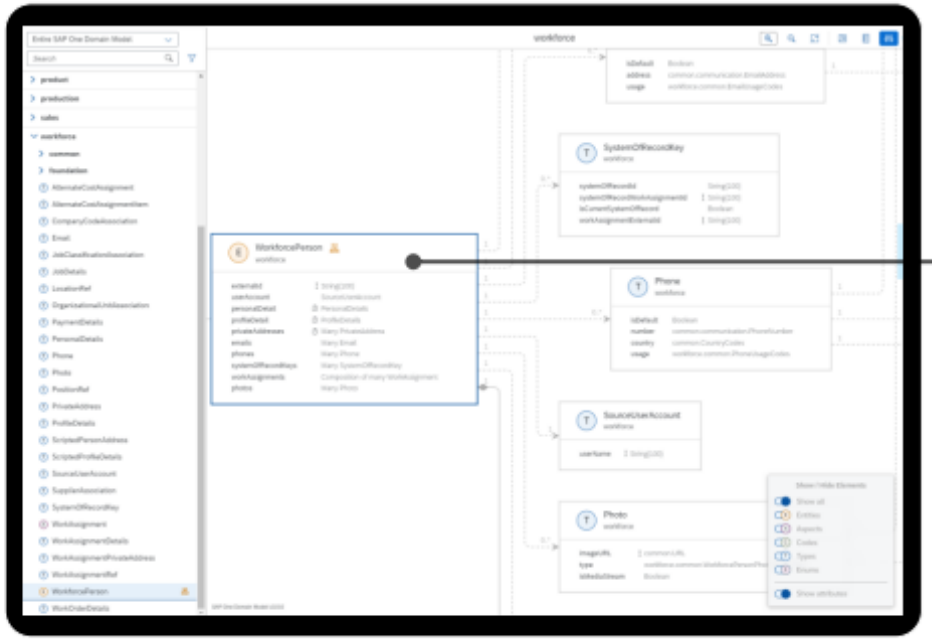
Central integration point

2

SAP One Domain Model as lingua franca for exchanging data

3

Out of the box integrations provided to cloud solutions from SAP



ODM

WorkforcePerson

externalID: String(100) not null
userAccount: SourceUserAccount
personalDetail: PersonalDetails
profileDetail: ProfileDetails
privateAddresses: Many PrivateAddress
emails: Many Email
phones: Many Phone
systemOfRecordsKeys: Many SystemOfRecordKey
workAssignments: Composition of many WorkAssignment
photos: Many Photo

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SAP Master Data Integration

1

Central integration point

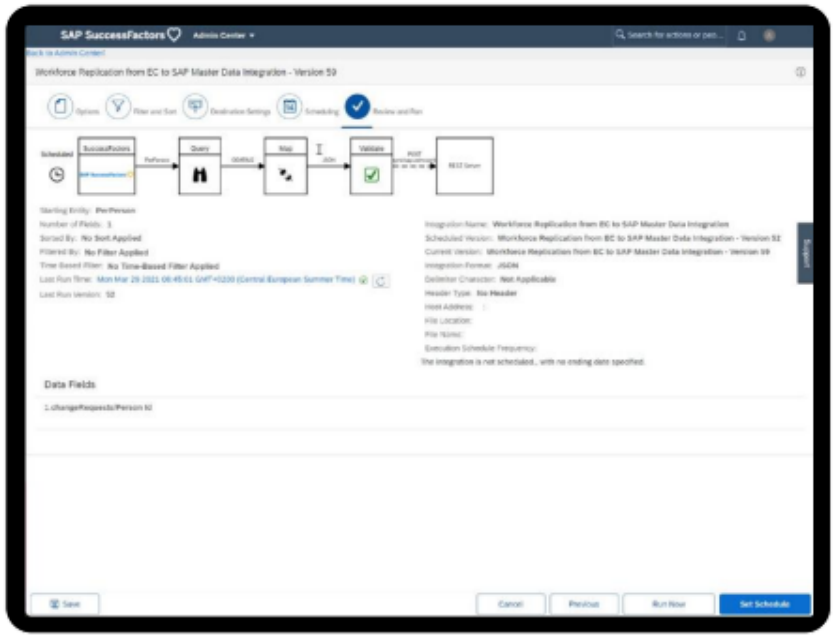
2

SAP One Domain Model as lingua franca for exchanging data

3

Out of the box integrations provided to cloud solutions from SAP

Step toward fully managed SAP scenarios



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SAP Master Data Integration: scenario’s

- S/4HANA Cloud

Master Data-Specific Information

- Bank Replication
- Company Code Replication
- Cost Center Replication
- Exchange Rate Replication
- Plant Replication
- Product Replication
- Product Group Replication
- Project Controlling Object Replication
- Public Sector Management Data Replication
- Purchasing Organization Replication
- Purchasing Group Replication
- Technical Object Replication
- Workforce Person Replication

- SuccessFactors

Scenario: Integrating Workforce from Employee Central to SAP Master Data Integration.

Scenario: Integrating Cost Center from SAP Master Data Integration to Employee Central

Scenario: Integrating Job Classification from Employee Central to SAP Master Data Integration.

Scenario: Integrating Organizational Units from Employee Central to SAP Master Data Integration.

Scenario: Integrating Bank from SAP Master Data Integration to Employee Central.

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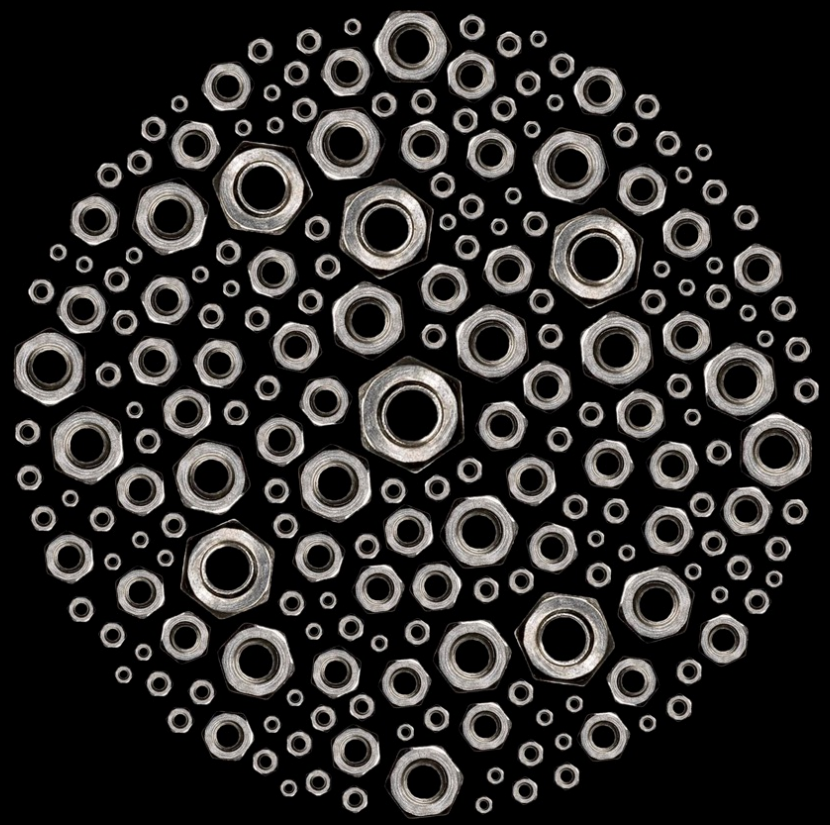
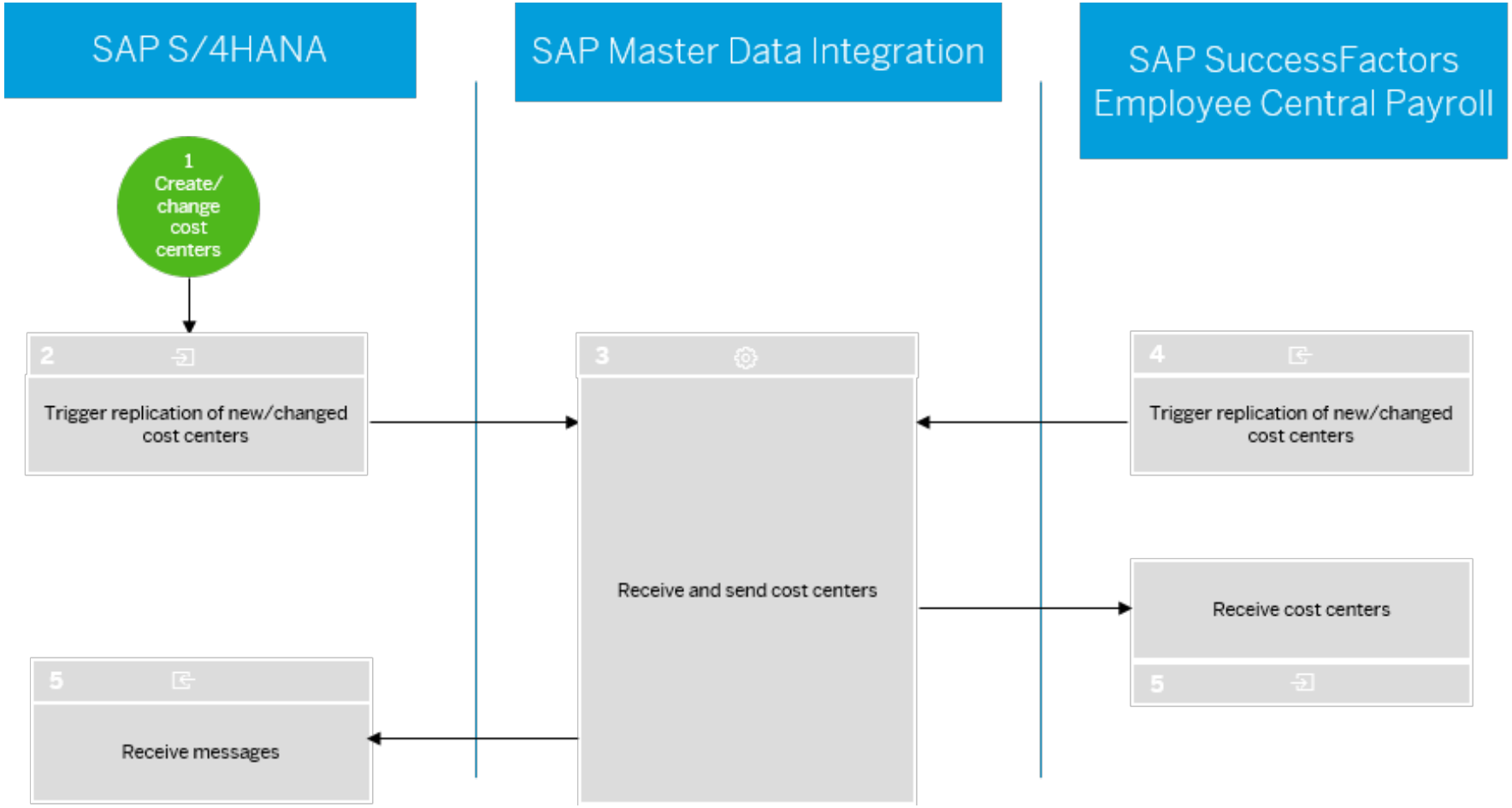


SAP Master Data Integration
Demo Scenario - Setup

SAP Master Data Integration: demo scenario

- Cost Center integration between S/4HANA Cloud and SuccessFactors
  - [Scenario: Integrating Cost Center from SAP Master Data Integration to Employee Central]



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SAP Master Data Integration: demo scenario

- Setup instructions: S/4HANA Cloud: link

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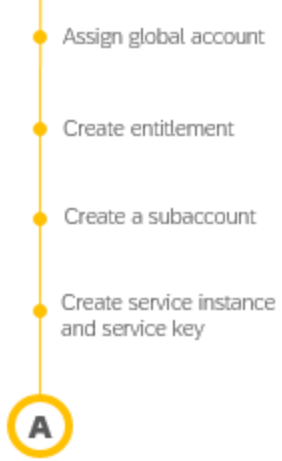
SAP Master Data Integration: demo scenario

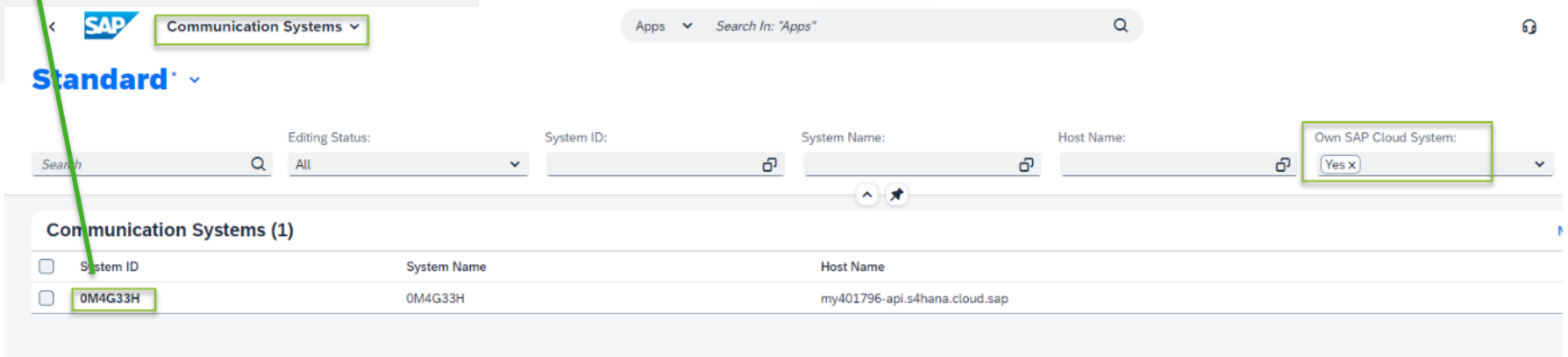


- Link BTP
  - Create MDI instances (1 for each system)

MasterDataIntegration	Master Data Integration	sap-integration
MDI S/4 Hana Demo my401796	Master Data Integration	sap-integration

  - Contains identification of the application, type of application and scope

```
{
    "application": "s4",
    "businessSystemId": "0M4G33H",
    "enableTenantDeletion": false,
    "writePermissions": [
        {
            "entityType": "sap.odm.finance.costobject.CostCenter"
        }
    ]
}
```



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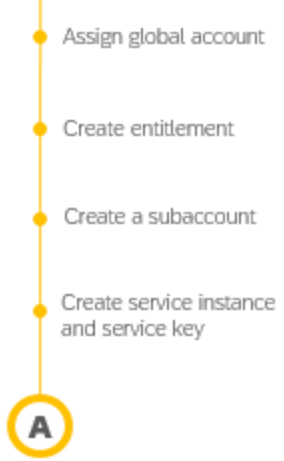
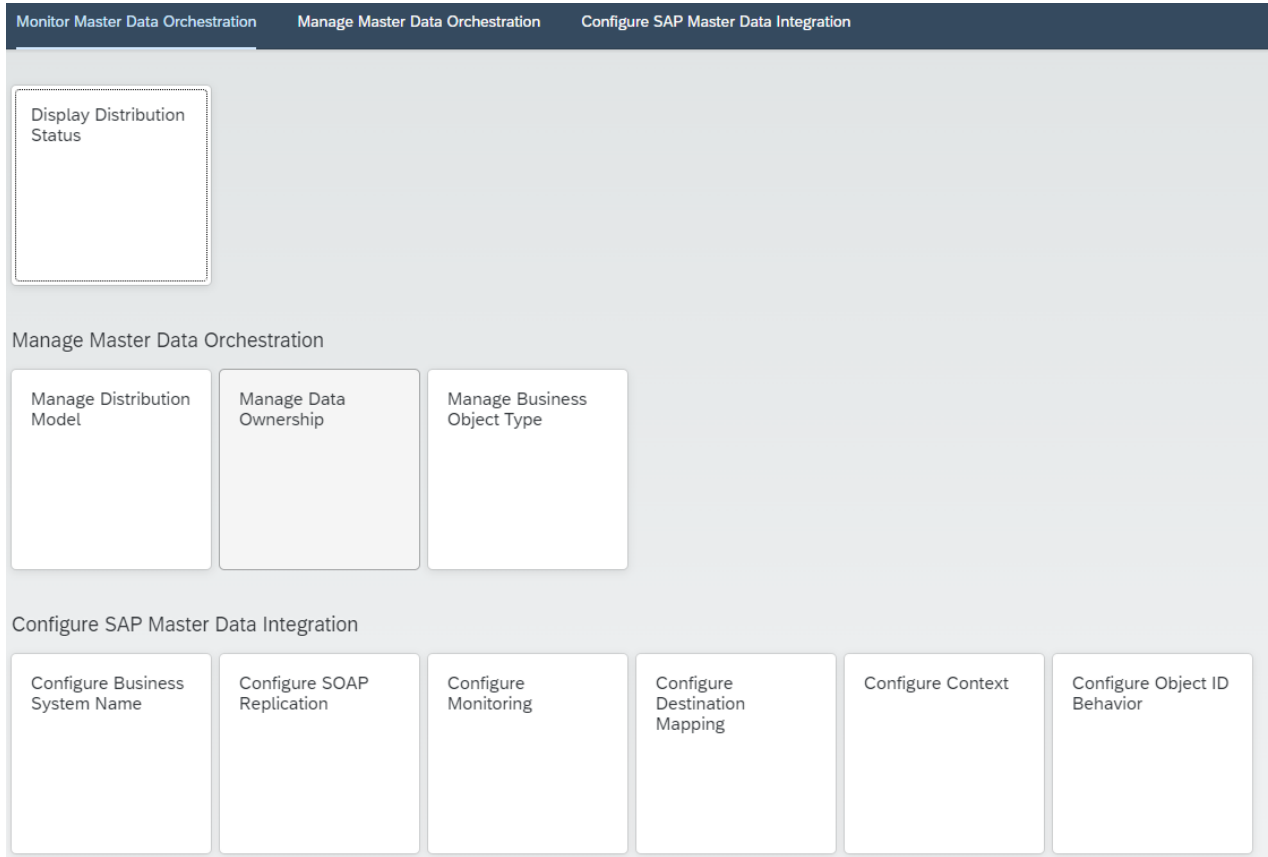
SAP Master Data Integration: demo scenario: setup BTP



- Link BTP
  - Subscribe to MDO (1 for all systems)

Master Data Integration (Orchestration)	standard

    - Creation of distribution models
    - Manage business object types + custom fields
    - Manage data ownership + sensitivity (read logging)



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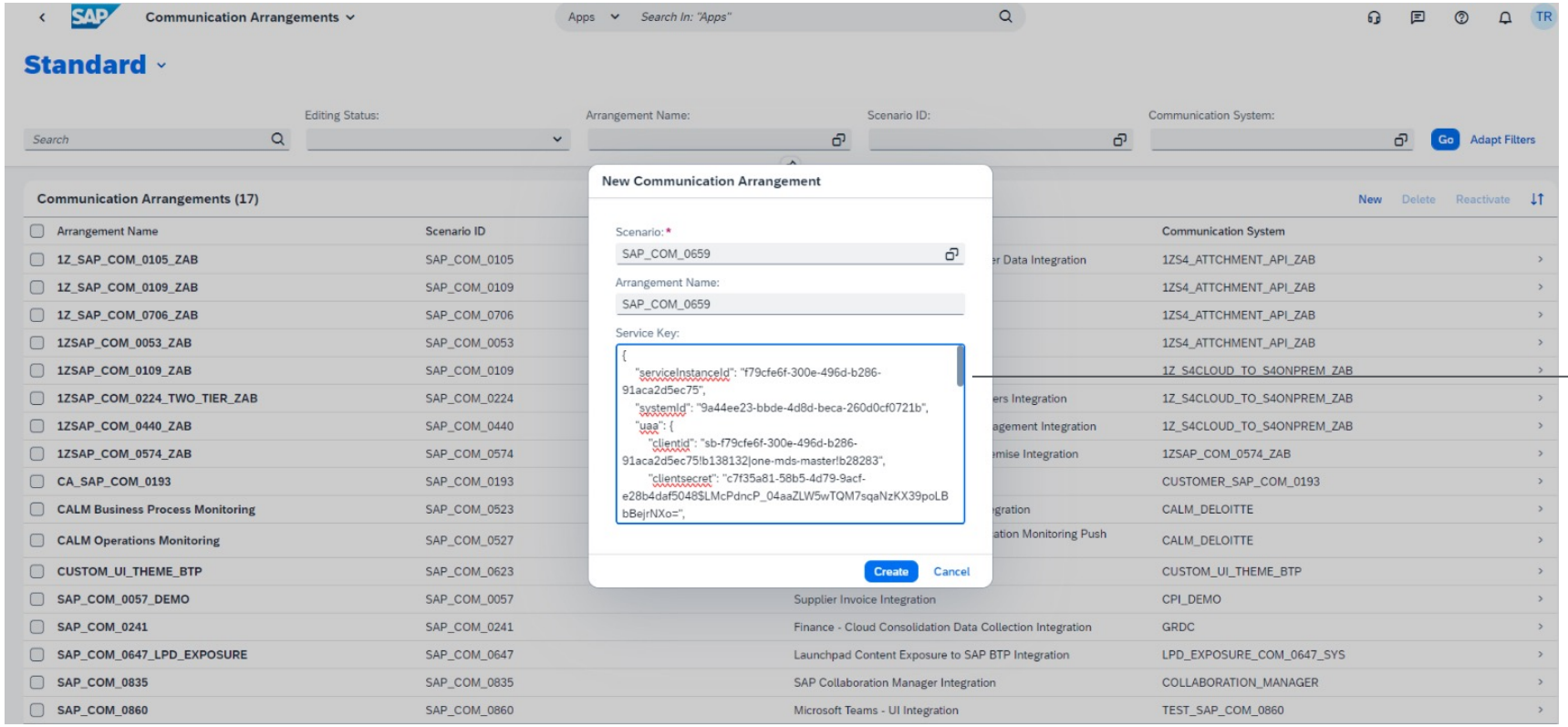
SAP Master Data Integration: demo scenario: setup S/4

- Link S/4HANA Cloud
  - Create communication arrangements with MDI and MDO
    - Link with MDI (0659)



Create communication arrangement with service key from BTP MDI instance

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SAP Master Data Integration: demo scenario: setup S/4

- Link S/4HANA Cloud
  - Create communication arrangements with MDI and MDO
    - Link with MDO (0594)

SAP_COM_0594

Scenario ID: SAP_COM_0594
Scenario: SAP Master Data Orchestration Integration

Draft Last Changed By: Tom Rottiers
Draft Last Changed On: 05.09.2023, 10:49:33

Editing Status: Draft

Common Data

Arrangement Name: SAP_COM_0594

Communication System:* New

Own SAP Cloud System: 0M4G33H

API-URL: https://my401796-api.s4hana.cloud.sap

Inbound Communication

Supported Authentication Methods

User Name:*

Authentication Method:

Inbound Services

Service	Application Protocol	Service URL / Service Interface	WSDL/Service Metadata	Additional Properties
MDO_REPLICATION_0001 (Deprecated) Deprecated with release 2308	OData V2	https://my401796-api.s4hana.cloud.sap/sap/opu/odata/sap/MDO_REPLICATION		
MDO_ORCHESTRATIONADMIN_0001 (Deprecated) Deprecated with release 2308	OData V2	https://my401796-api.s4hana.cloud.sap/sap/opu/odata/sap/MDO_ORCHESTRATIONADMIN		
MDO: Distribution Administration	OData V4	https://my401796-api.s4hana.cloud.sap/sap/opu/odata4/sap/mdo_distributionadmin/srvd_a2x/sap/distributionadmin/0001/		
MDO: Replicate Distribution	OData V4	https://my401796-api.s4hana.cloud.sap/sap/opu/odata4/sap/mdo_replicatedistribution/srvd_a2x/sap/replicatedist		

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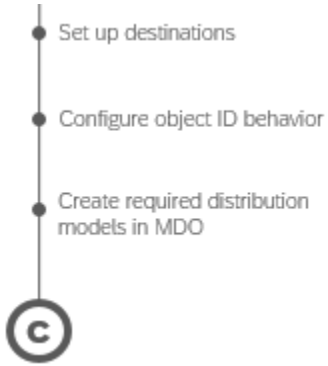
SAP Master Data Integration: demo scenario: configure MDO



- Link MDO
  - Setup destination on BTP toward S/4 (used in the distribution model)

Destination Configuration

Name: *	S4-Cloud-my401796
Type:	HTTP
Description:	MDI/MDO
URL: *	https://my401796-api.s4hana.cloud.sap/sap/opu/odata4/sa...
Proxy Type:	Internet
Authentication:	BasicAuthentication
User: *	MDO
Password:	••••••••

Additional Properties

MDIInstanceId	9a44ee23-bbde-4d8d-b...
MDOBusiness...	S4-Cloud-my401796
MDOConsumer	true
MDOProvider	true

☑ Use default JDK truststore

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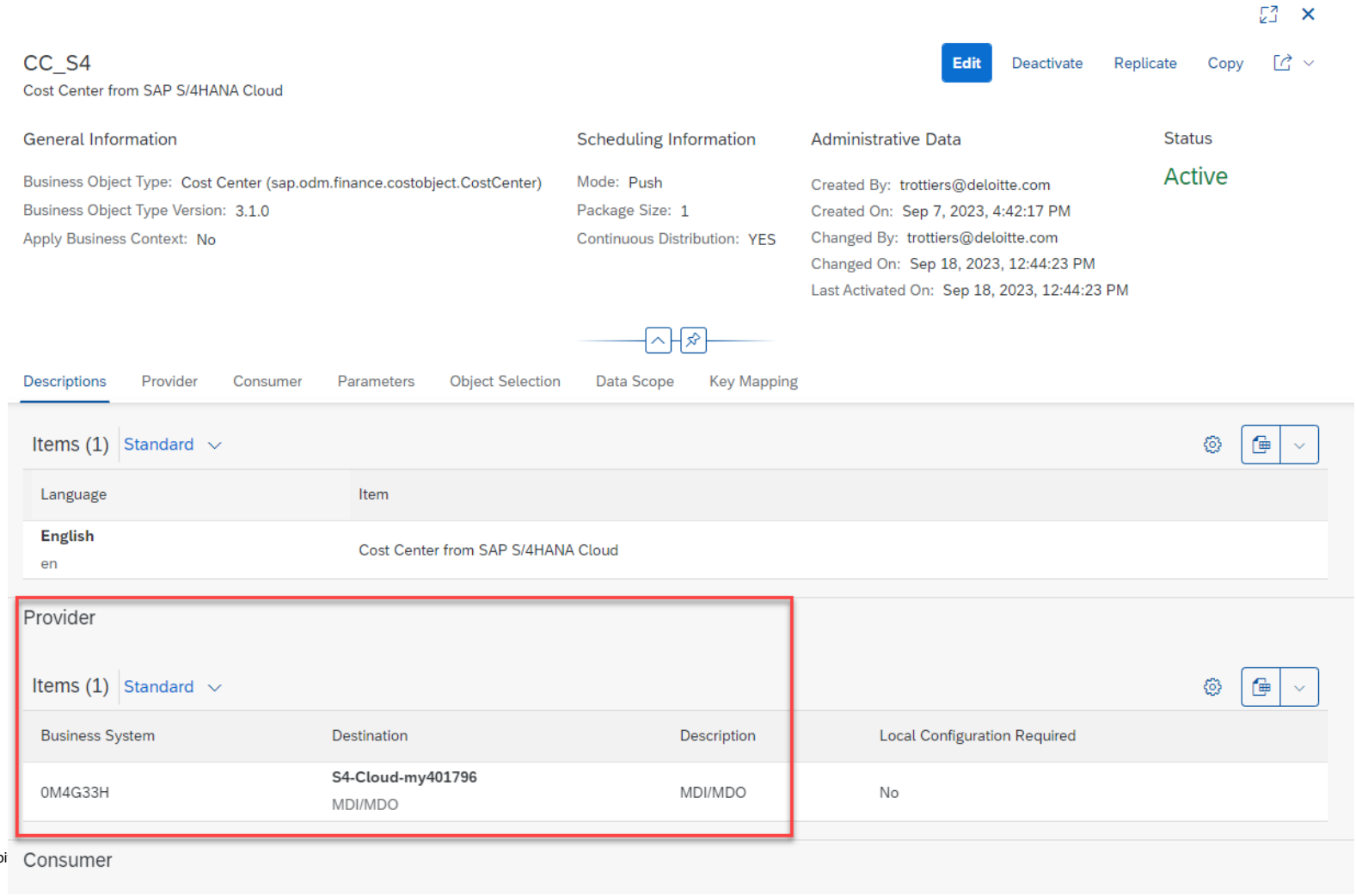
SAP Master Data Integration: demo scenario: configure MDO

- Link MDO
  - Creation distribution model in MDO



Set up destinations

Configure object ID behavior

Create required distribution models in MDO

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SAP Master Data Integration: demo scenario: configure MDO

- Set up destinations
- Configure object ID behavior
- Create required distribution models in MDO

- Link MDO
  - Creation distribution model in MDO + activate

Consumer

Items (1) Standard

Business System	Destination	Description	Local Configuration Required	Application Category	Business Contexts
MDI	MDI (SAP Master Data Integration)	SAP Master Data Integration	No		–

Parameters

Business Object Type	**Model Parameters**	**Scheduling Information**
Business Object Type: Cost Center (sap.odm.finance.costobject.CostCenter)	Package Size: 1	Continuous Distribution: YES
Business Object Type Version: 3.1.0	Mode: Push	
API: MDI_CHANGE_COST_CENTER		

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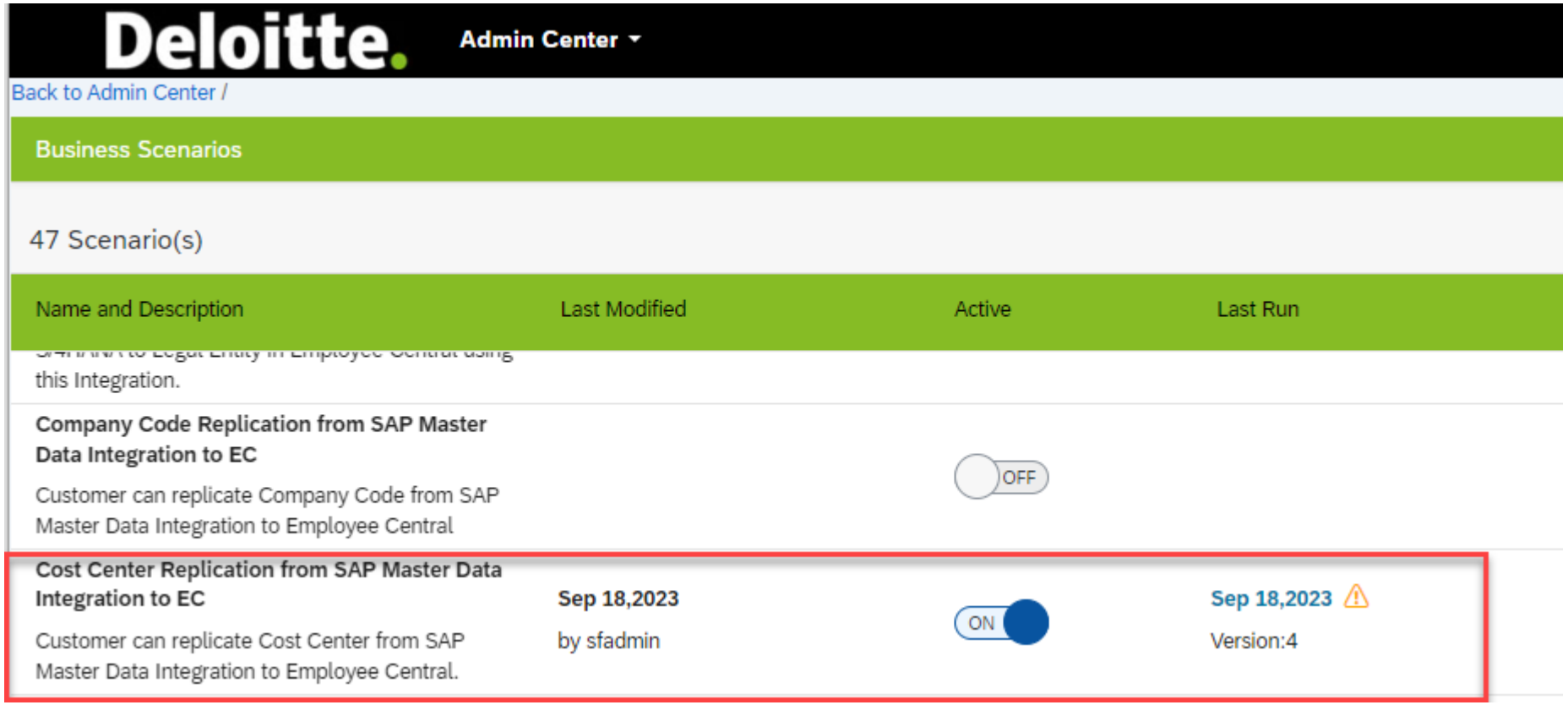
SAP Master Data Integration: demo scenario: configure SF

- Link SF
  - Activation of business scenario



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SAP Master Data Integration: demo scenario: configure SF

- Link SF
  - Activation of business scenario

Cost Center Replication from SAP Master Data Integration to EC - Version 4

Options | Source Settings | Scheduling | Review and Run

REST Server Settings

Connection Name: SAP Master Data Integration

REST API URL *: https://one-mds.cfapps.eu10.hana.ondemand.com

Authentication Type: OAuth

OAuth Configurations: MasterDataIntegration

Click to manage OAuth Configurations

Delta Token: * K2433859274u3uAu5u...

REST Operation Settings

Operation Name:

Operation URI*: /v0/odm/3.0.0/log/sap.odm.finance.costobject.CostCenter

HTTP Method: GET

Payload Format: JSON

Source Page Size: 1000

☑ Enable diagnostics for Run Now

☐ Execute with exceptions

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SAP Master Data Integration: demo scenario: setup monitoring in CALM ☹

https://deloitte-consulting-advisory-cvba-cloudalm.eu10.alm.cloud.sap/launchpad

SAP Cloud ALM Application URL

SAP Cloud ALM Application | Monitoring Service Details

SAP Cloud ALM Application URL:
https://deloitte-consulting-advisory-cvba-cloudalm.eu10.alm.cloud.sap/launchpad

Error

Wrong literal format for literal: '[undefined]'.

OK

3d-cloudalm!b16907

BJ

To Rottiers, Tom

You forwarded this message on 20/09/2023 11:29.

Hello Tom,

Thank you for the problem report. Sorry about this... The error you shared is already known to us and is caused by a bug in MDO. We are working on a solution to this problem, and will aim to get it deployed ASAP. However, we cannot provision a fix to the productive landscapes before September 26th for

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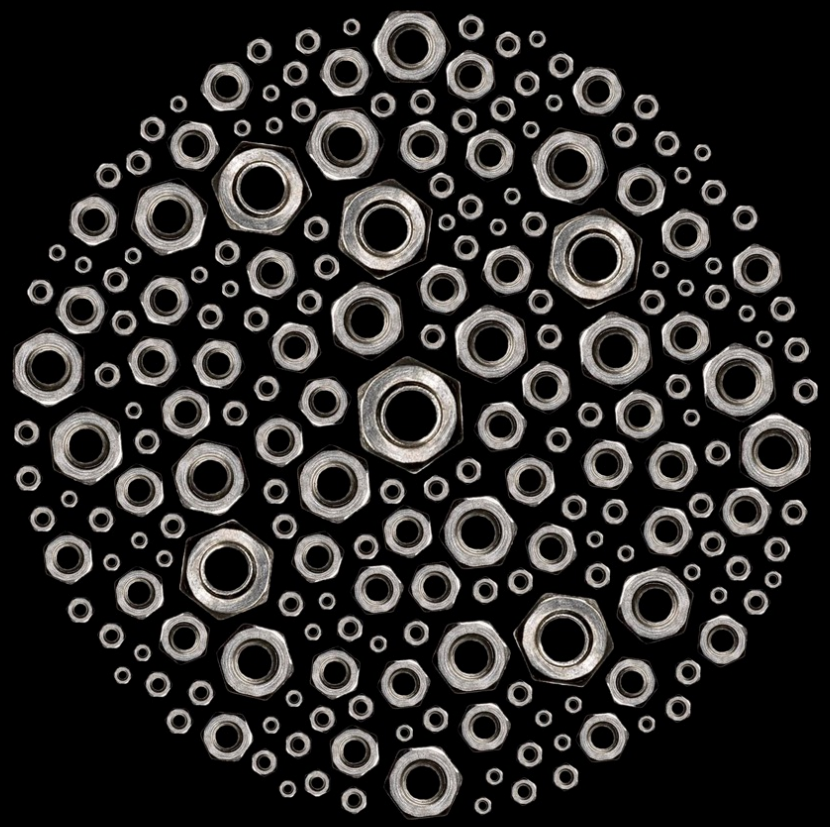
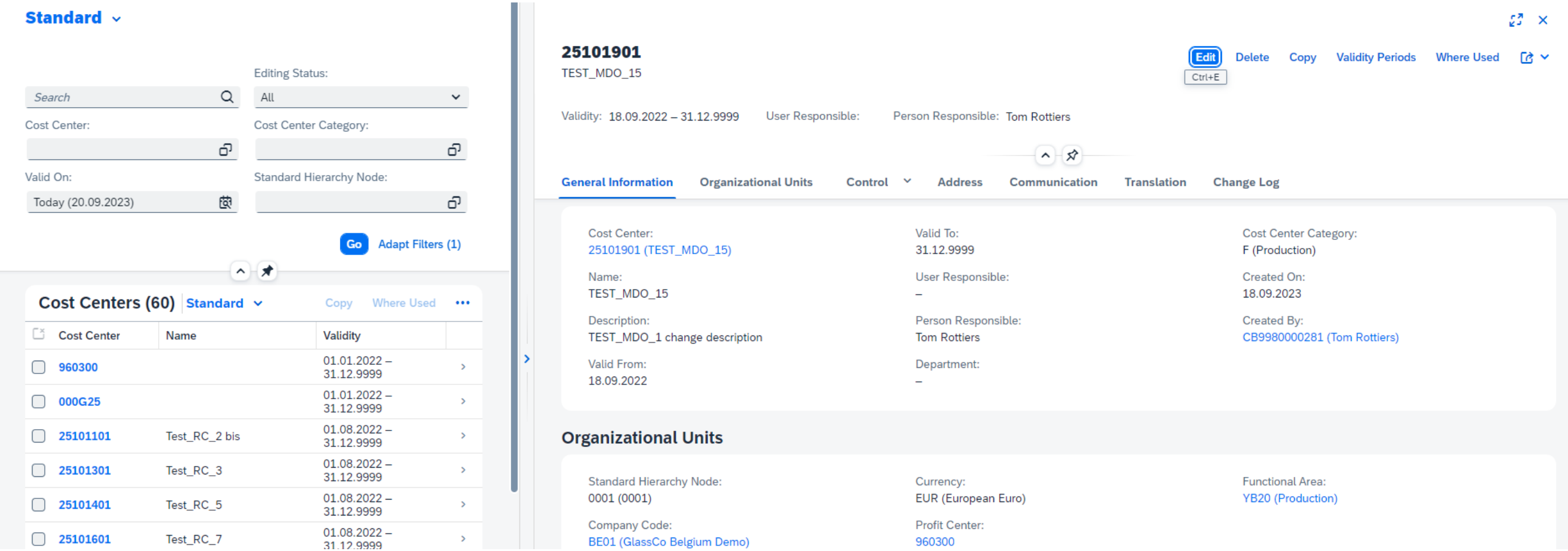


SAP Master Data Integration: demo time

SAP Master Data Integration: demo

- Link S/4HANA Cloud
- ‘Manage cost centers’ + change cost center + save



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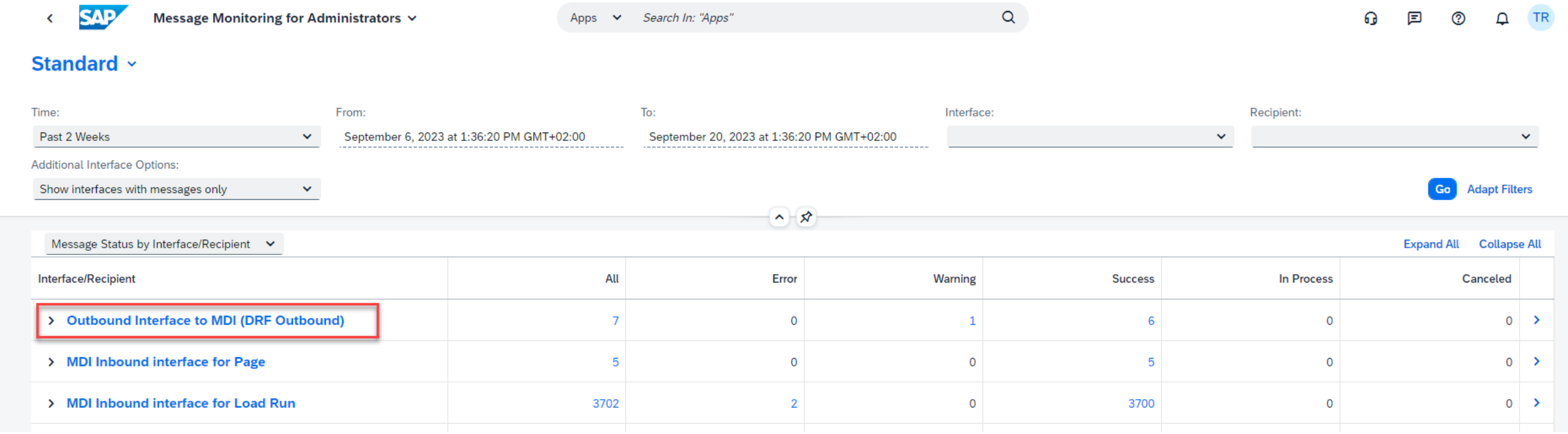
SAP Master Data Integration: demo

- Link S/4HANA Cloud
- Verify change and replication status (Message monitoring for administrators)



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SAP Master Data Integration: demo

- Link S/4HANA Cloud
- Verify change and replication status (Message monitoring for administrators)

Standard

Filtered By (4): From : September 6, 2023 at 11:37:09 AM GMTZ; To : September 20, 2023 at 11:37:09 AM GMTZ; Interface : Outbound Interface to MDI (DRF Outbound); Recipient : /MDI-MDI_ALL

Messages (7)

Status	Processing Date
Warning	20.09.2023, 11:34:59
Message ID:	6792C4FA43D81EDE95F5375AD1867E9E
Success	20.09.2023, 11:34:58
Message ID:	6792C4FA43D81EDE95F5373F24237E9E
Success	18.09.2023, 12:29:32
Message ID:	6792C4FA43D81EDE95C3E0A32E3D7E9B
Success	18.09.2023, 12:23:45
Message ID:	6792C4FA43D81EDE95C3C6B431E9BE9B
Success	18.09.2023, 10:49:42
Message ID:	6792C4FA43D81EDE95C2223F69F0FE9B
Success	18.09.2023, 10:42:48
Message ID:	6792C4FA43D81EDE95C203708F0CBE9B
Success	18.09.2023, 10:32:22
Message ID:	6792C4FA43D81EDE95C1D4D456867E9B

6792C4FA43D81EDE95F5375AD1867E9E

Status
Warning

Processor: Not Assigned
Process Status:
Comments: 0

General Information | Log Details | Data Content

Key Information

Bus. Obj. Type:	DRF_0081
Run Id:	000000000000007
BO Instance ID:	6792c4fa-43d8-1ede-95c2-03708f0bde9b
Local ID:	BE010025101901
BO Display ID:	

Log Details

All 0 1 0 0

Log Message
Objects wait to be sent (run id 7)

Log Message

MDI has Accepted the Change Request

SAP Master Data Integration: demo

- Link SuccessFactors to run import processed (scheduled)

	Event Name	Event Description	Event Type	Created Date
	endJob	End execution of job [jobExecutionId=1307315623]	END	Mon Sep 18 2023 14:30:43 GMT+0200 (Central European Summer Time)
	Replication Status (WARN)	Display Name: BE01/0025101901/0M4G33H, Error details: Time slice start date 2022-09-18 Legal entity "BE01" of cost center: 0025101901 does not exist in EC, External Code: BE01/0025101901/0M4G33H, OID: 6792c4fa-43d8-1ede-95c2-03708f0bde9b, Version Id: 019c2c79-0798-40eb-a273-dadd5a8e0e93, Event Type: updated	WARNING	Mon Sep 18 2023 14:30:42 GMT+0200 (Central European Summer Time)
	Create/Update/Deactivate	Received: 1, Success count: 0, Warning count: 1, Error count: 0	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:42 GMT+0200 (Central European Summer Time)
	Diagnostic Data	Attached is raw data read from source	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:42 GMT+0200 (Central European Summer Time)
	Page 1	Number of records read :1. Total records read so far:1	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:42 GMT+0200 (Central European Summer Time)
	Integration Process Info	Successfully generated authentication token from URL https://poc-integration-suite-sle12aqb.authentication.eu10.hana.ondemand.com/oauth/token	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:41 GMT+0200 (Central European Summer Time)
	Integration Center Info	Couldn't perform Read Access Logging. Ensure you have also enabled the Read Access Logging switch from Admin Center > Manage Audit Configuration > Read Audit.	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:41 GMT+0200 (Central European Summer Time)
	Query	mainEntityName : FOCostCenter, entitySelect : , entityFilter : null, entityOrderBy : null, entityExpand : null, globalAsOfDate : null, globalFromDate : null, globalToDate : null, lastSuccessfulRunTime : Mon Sep 18 14:28:05 CEST 2023	SUMMARY_SO_FAR	Mon Sep 18 2023 14:30:40 GMT+0200 (Central European Summer Time)
	startJob	Start execution of job [jobExecutionId=1307315623]	START	Mon Sep 18 2023 14:30:40 GMT+0200 (Central European Summer Time)

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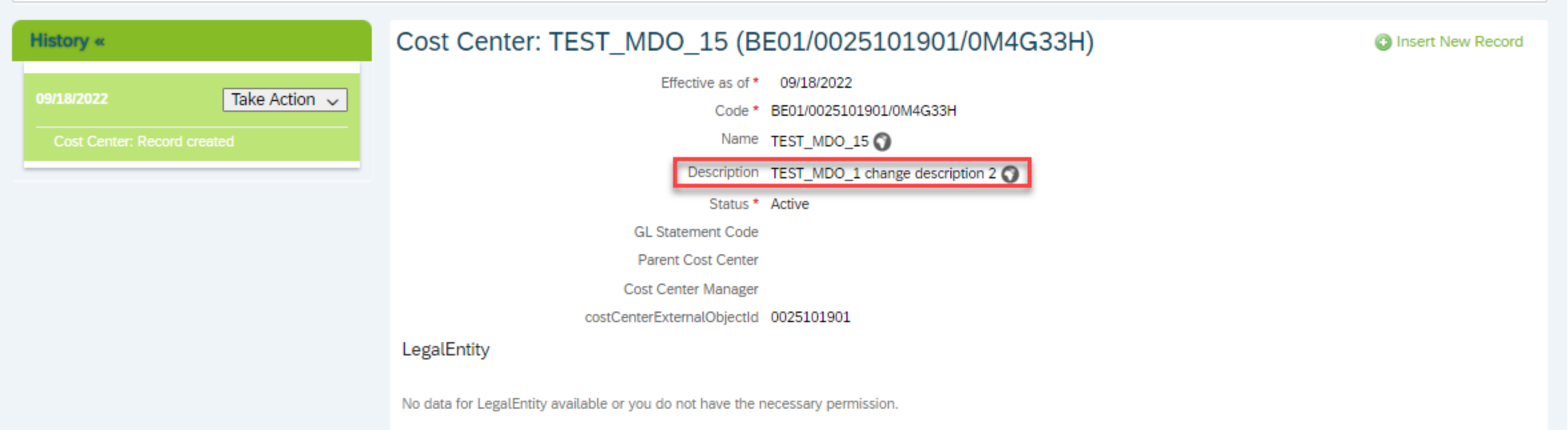
SAP Master Data Integration: demo

- Link SuccessFactors to run import processed (scheduled)



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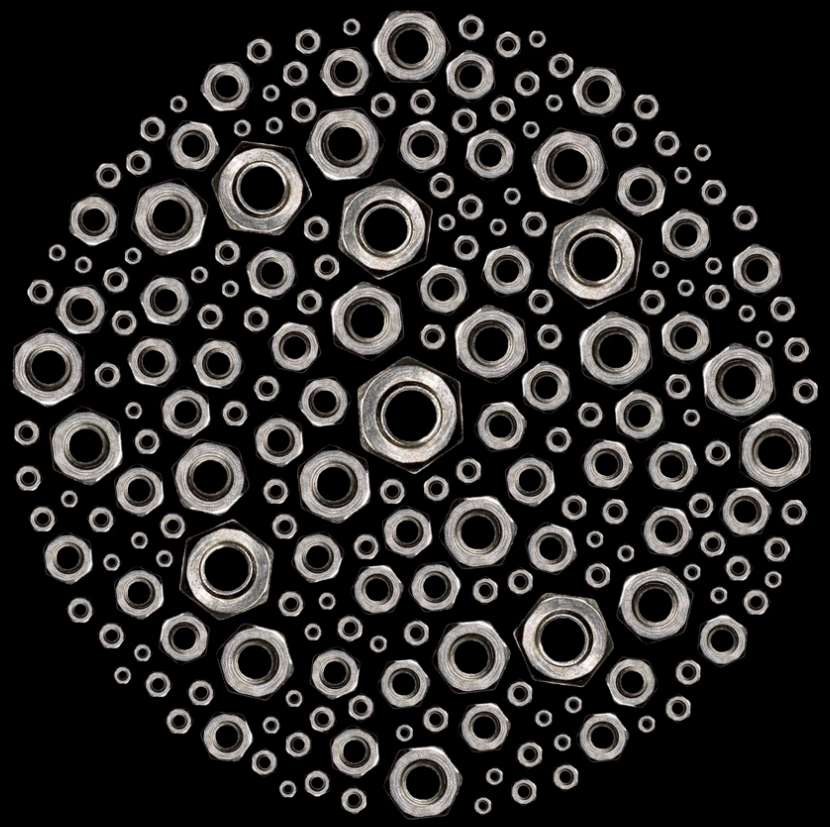


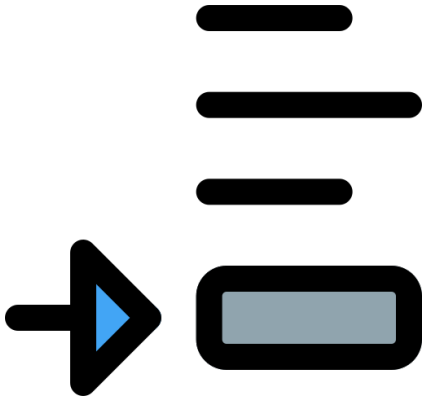
Conclusion

SAP Master Data Integration: demo

- Tool has a lot of potential and responds to the need of many companies (canonical representation of business data)
- Value increases with number of SAP LoB solutions
- Not yet 100% stable (creation of distribution model had some hiccups + monitoring bug)
- Monitoring is key → CALM would provide the centralized monitoring component

Still to test:

- Employee data to S/4HANA Cloud (BP)
- Monitoring through CALM
- Load



MDI is here to stay and it's the right time to start learning, experimenting and evaluating!!

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SAP Master Data Integration: resources

- SAP Master Data Integration in the Intelligent Enterprise | openSAP
- MDI | SAP Blogs
- https://help.sap.com/docs/SAP_MASTER_DATA_INTEGRATION
- SAP Discovery Center - Master Data Integration (cloud.sap)
- https://roadmaps.sap.com/board?PRODUCT=73555000100800002242
- Integration | SAP Help Portal (MDI supported applications)
- Guided Answers (sap.com)

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